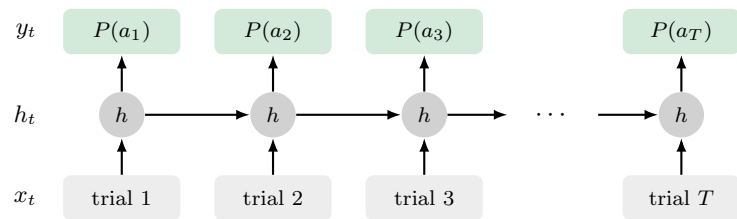


Many-to-many: every step gets an input x_t and produces an output y_t ; the loss is summed over the steps that carry a label

1. Trial-level tiny RNN (one step = one trial; the sequence is a whole session)



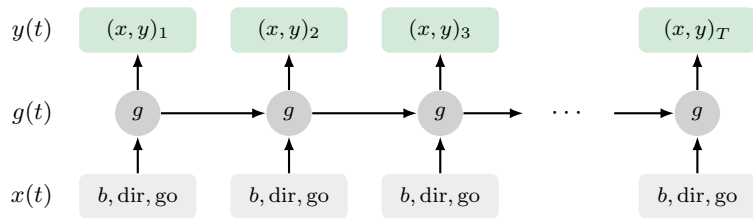
$y_t = P(a_t)$: probability of Short / Mid / Long on trial t compared with the animal's choice a_t (cross-entropy); also the decision time, execution time and takeoff heading (Section 8)

h_t : $d = 1-4$ variables carried from trial to trial

$x_t = (a_{t-1}, s_{t-1}, r_{t-1}, s_t, \text{cue}_t)$: previous choice, previous bar, previous outcome, current bar, rule

Every trial has a label, so every step contributes to the loss. Sequence length $T = 240-465$ trials.

2. Motor (joystick) network (one step = 10 ms; one trial, ≈ 335 steps)



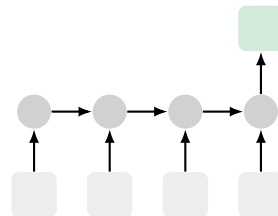
$y(t) = (x, y)(t)$: cursor position at every step, compared with the measured trajectory (MSE on valid steps)

$g(t)$: GRU state, M units

$x(t) = (\mathbf{b}, \text{dir}, \text{go}(t))$: $\mathbf{b}$ is trial 1's own read-out (a function of h_t), **held constant** across the trial's steps; chosen direction one-hot; go

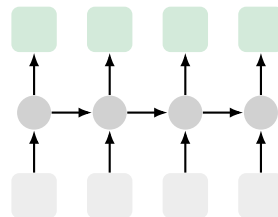
A measured position at every step, so every valid step contributes. This is a purer many-to-many than row 1: no output is masked.

Many-to-one



reads the whole sequence, one label at the end (e.g. the section 1.10 decoder)

Many-to-many (aligned)



one output per step; a mask decides which steps enter the loss (row 1 only)